



Computational complexity and exact techniques for scheduling problems with job prohibitions

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Abstract

The scheduling problems with job prohibitions are investigated. Here we are given a system of job subsets for positions of machines. For each position of a machine the subset indicates jobs, which can be performed in this position. The considered problems arise in the case of technological requisitions in production systems and multi-processor computer systems, where the order of job execution is influenced by fixed routes and structural constraints. At the same time, such problems play an important role in evolutionary algorithms, when the optimal recombination problem is solved in a crossover operator, and other iterative techniques based on solving a series of subproblems with job prohibitions. We analyze the computational complexity of the problem for various regular criteria in single-stage statements and multi-stage systems. We also discuss a general method to proving the NP-hardness of the specified class of scheduling problems using the polynomial reduction of the Ordered Partition Problem to the decision version of the problem, when an instance with the non-idle property is constructed. A solving approach based on the enumeration scheme and mixed integer linear programming models is proposed. This approach allows us to show that “almost all” instances are polynomially solvable, and it is important for evolutionary computation and other techniques based on enumerating partially mapped solutions.

Keywords Scheduling · Prohibition · Algorithm · Complexity

1 Introduction

Scheduling problems on permutations with job prohibitions are considered in the following statement. Let $\mathcal{J}$, $|\mathcal{J}| = n$, be the set of jobs and $\mathcal{M}$, $|\mathcal{M}| = m$, be the set of machines. We investigate single-stage systems (Brucker, 2007; Graham et al., 1979), where each job consists of a unique operation, and multi-stage systems of flow-shop, job-shop and open-shop types (Sotskov & Shakhlevich, 1995; Tanaev et al., 1989), where each job consists of several operations. Denote by p_{ij} the duration (processing time) of operation i of job j . The set of positions $N_l = \{1, \dots, n_l\}$ is given for each machine $l \in \mathcal{M}$, here jobs (or their operations) can be placed. A system of job subsets $X = \{X^{i,l}\}_{i \in \mathcal{M}, l \in N_l}$ is also given. $X^{i,l}$ is the subset of jobs (operations), which can be performed in position $i \in N_l$ of machine $l \in \mathcal{M}$. As a result, we have prohibitions on executing jobs in some positions

of machines. The goal is to assign jobs (or their operations) to positions of machines in accordance with the system of job subsets X so that a polynomially computable regular criterion has the minimum value. Such scheduling problems with additional constraints on executing jobs in positions of machines are called in the literature as problems with prohibitions (Serdyukov, 1978) or problems with requisitions (Eremeev & Kovalenko, 2017).

It is evident that we have the classic scheduling problem, when $|X^{i,l}| = n$ for all positions of machines. At the same time, the problem is polynomially solvable in many cases, when $|X^{i,l}| = 1$ for each position. We will analyze the computational complexity of the problem for various regular criteria and cardinality of subsets from X . Moreover, we will discuss a general approach to proving the NP-hardness of the specified class of problems using the reduction from the Ordered Partition Problem of a special form, when a family of instances with an optimal solution without machine idle time is constructed. We will present approaches for solving the problem when $|X^{i,l}| \leq 2$ and arbitrary.

Such problems have applications in computer systems and manufacturing plants, where prohibitions for performing of

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jobs in positions of machines arise due to technological constraints, fixed routes (Chernykh, 2016; Fisher, 1995) and working shifts (Ernst et al., 2004).

For example, consider the scheduling problem with routing (Chernykh, 2016). An undirected edge-weighted graph is given. Vertices correspond to nodes of the transportation network. Jobs are placed in nodes. The weights of the edges correspond to traveling times between nodes. A machine visits a node at most once. Jobs are performed by machines in nodes, where they placed. We may consider models in which jobs have precedence constraints, multiple stages and other relations. The goal is to minimize some regular criterion. Such problems can be solved in two stages. Initially we find routes of machines and then schedule jobs in positions of routes (Chernykh, 2016; Chernykh et al., 2013). However, when routes of machines are fixed, then we obtain a scheduling problem with job prohibitions defined by nodes visited by machines in positions of routes (Chernykh, 2016; Chernykh et al., 2013; Fisher, 1995).

At the same time, such problems play the important role in evolutionary algorithms, where the optimal recombination problem is solved in a crossover operator. Parent solutions exchange their properties to generate offspring solutions in crossover operators of evolutionary computation (Balas & Niehaus, 1998; Radcliffe, 1994; Yagiura & Ibaraki, 1996). Given two parent solutions, the Optimal Recombination Problem consists in finding the best possible offspring that can emerge as a result of a crossover where a value of each solution component is copied from one of the values of this component in the parent solutions. Such recombination operators are called (strictly) gene transmitting (Radcliffe, 1994).

The optimal recombination problem in the case of single machine is formulated as follows (Balas & Niehaus, 1998; Ereemeev & Kovalenko, 2014):

Given two parent permutations of jobs $\pi^1 = (\pi_1^1, \dots, \pi_n^1)$ and $\pi^2 = (\pi_1^2, \dots, \pi_n^2)$. It is required to find an offspring permutation $\pi' = (\pi_1', \dots, \pi_n')$ such that

- (I) $\pi_i' = \pi_i^1$ or $\pi_i' = \pi_i^2$ for all $i = 1, \dots, n$;
- (II) π' has the optimum value of the given objective over all permutations satisfying condition (I).

Then jobs π_i^1 and π_i^2 compose subset $X^{i,1}$ for position i of the machine. The optimal recombination problem is formulated in the appropriate way for the case of several machines.

Investigation of the computational complexity of the optimal recombination for the classic routing problems and Boolean programming problems, permutation flow-shop and single-machine scheduling with changeovers has been provided in Ereemeev and Kovalenko (2012); Kovalenko (2016); Ereemeev (1999). For solving such problems, various approaches are used, such as dynamic programming (Yagiura & Ibaraki, 1996), enumeration techniques from the graph the-

ory (Ereemeev & Kovalenko, 2020; Kovalenko, 2016; Tinos et al., 2020; Uno, 1997) with recursion or decomposition, and others. As we can see from the computational evaluations (see, e.g., Balas & Niehaus, 1998; Chicano et al., 2019; Ereemeev & Kovalenko, 2016; 2020; Tinos et al., 2020; Yagiura & Ibaraki, 1996; Zakharova, 2023a; 2023b), the optimal recombination allows to improve the results of evolutionary algorithms with randomized operators. Here restricted versions of exact algorithms and greedy methods are used for solving NP-hard optimal recombination problems in the evolutionary search.

Serdyukov (1978) proposed an exact algorithm for the TSP problem with vertex prohibitions based on enumerating of perfect matchings in the special bipartite graph. This algorithm was improved in Ereemeev and Kovalenko (2017). In Saadani et al. (2005), Kovalyov and Werner (2007), the two-machine flow shop scheduling problem with forbidden jobs in first or last position was investigated. The authors provided motivation from branch and bound algorithms for the no-wait variant of the flow-shop and proposed polynomial time algorithms.

There are known approximation algorithms for the routing open shop problem, where routes of machines are constructed at the first step, and schedules of jobs in nodes of routes are formed at the second step (Chernykh, 2016; Chernykh et al., 2013). However, the authors consider all jobs belonging to the same node as one job in their proofs. So, it is interesting to investigate job prohibitions for positions of machine permutations and search for possibility of improving procedures in this context.

The results of the current paper allow to provide new properties of scheduling problems with prohibitions, various criteria and constraints. These properties guarantee NP-hardness in the worst case and polynomial solvability in “almost all” cases for practically important instances. It places important role in evolutionary computation and other iterative methods, where sub-problems with prohibitions are solved at each iteration via MIP or enumerating based techniques.

As previous research we also mention list coloring-type problems. Some scheduling problems (multi-stage problems, problems with grouping machines by technologies, resource allocation problems) contain as subproblems coloring-type problems, which consist in deciding whether a graph is k -colorable or not for some natural number k . The well-known modifications are list-coloring, list incidentor coloring, μ -coloring and (λ, μ) -coloring (Bonomo et al., 2009). Here we have additional constraints on colors that can be used for coloring vertices or edges. The List chromatic number problem is a subproblem of the scheduling problem with single mode multi-processor jobs (correspond to vertices) and job prohibitions, and the List chromatic index problem is also can be interpreted as scheduling single mode two-

processor jobs (correspond to edges) with job prohibitions (Drozdowski, 2009). In the interpretation of list incidentor coloring as job-shop scheduling problems we will have a system X of subsets for two-stages jobs in the context of waiting time between operations (Pyatkin, 2008).

The paper has the following structure. Section 2 provides the general idea of proving NP-hardness of the single-machine scheduling problems with job prohibitions. This idea is realized on a series of problems. Moreover, an exact method and integer programming models are provided. Scheduling problems with job prohibitions, arising in multi-stage systems are investigated in Sect. 3. We prove NP-hardness of shop-problems with prohibitions and discuss approaches for solving. Finally we present the concluding remarks and open questions for further research.

2 Single machine system

In this section we consider the non-preemptive scheduling problem with job prohibitions on a single machine. Each job $j \in \mathcal{J}$ is characterized by release date r_j , due date d_j , duration (processing time) p_j and weight w_j . Preemptions of jobs are disallowed. It is required to find a sequence of jobs, placing at i -th position a job from the given subset X^i , $i = 1, \dots, n = |\mathcal{J}|$, and minimizing some regular criterion. Note that the early schedule corresponding to a permutation of jobs gives the best value of a regular criterion for this permutation.

Recall the classic notations: C_j is the completion time of job j ;

$T_j = \max\{0; C_j - d_j\}$ is the tardiness of job j ;

$L_j = C_j - d_j$ is the lateness of job j ;

$U_j = \chi_{C_j > d_j}$ is the indicator that job j is tardy:

$$\chi_{C_j > d_j} = \begin{cases} 1, & \text{if } C_j > d_j, \\ 0 & \text{otherwise, } j = 1, \dots, n. \end{cases}$$

We consider the following problems:

- $1|r_j| \sum_j C_j$ (the total completion time is minimized);
- $1|r_j| \sum_j U_j$ (the number of tardy jobs is minimized);
- $1|r_j| \sum_j T_j$ (the total tardiness is minimized);
- $1|C_j \leq d_j| \sum_j w_j C_j$ (the weighted total completion time is minimized); $1|r_j| \sum_j w_j U_j$ (the weighted number of tardy jobs is minimized);
- $1|r_j| \sum_j w_j T_j$ (the weighted total tardiness is minimized);
- $1|l_{ij}| C_{\max} = \max_j C_j$ (the makespan, here l_{ij} is the sequence dependent setup time between jobs i and j);
- $2|d_j = d|\gamma, \gamma \in \{C_{\max}, L_{\max}, \sum_j U_j, \sum_j T_j\}$ (two machines)

with additional constraint on job prohibitions.

Note that problems $1|\sum_j w_j U_j, 1|C_j \leq d_j| \sum_j w_j C_j$ are weakly NP-hard, but problems $1|r_j| \sum_j C_j, 1|r_j| L_{\max},$

$1|r_j| \sum_j U_j, 1|r_j| \sum_j T_j, 1|l_{ij}| C_{\max}$ are strongly NP-hard in the statement without prohibitions (Brucker, 2007; Lenstra et al., 1977).

2.1 NP-hardness

For proving NP-hardness of the considered problems we will use a polynomial reduction of the following NP-hard Ordered Partition Problem: Given an ordered set $A = \{a_1, a_2, \dots, a_{2n_0}\}$ and weight e_i of each element $a_i \in A$ such that $\sum_{a_i \in A} e_i = 2E$ and $e_i < e_{i+1}$, $i = 1, \dots, 2n_0 - 1$. The question is to decide whether A can be partitioned into two subsets A_1 and A_2 so that

$$\sum_{a_i \in A_1} e_i = \sum_{a_i \in A_2} e_i = E,$$

and subset A_1 contains only one element from each pair a_{2i-1}, a_{2i} , $i = 1, \dots, n_0$. Note that $|A_1| = |A_2| = n_0$ in the case of existence.

Proposition 1 *The Ordered Partition Problem is NP-hard.*

Proof Consider the classic Partition Problem (Garey & Johnson, 1979) with the set of elements $B = \{b_1, b_2, \dots, b_{n_0}\}$. The question is to decide whether B can be partitioned into two subsets B_1 and B_2 so that

$$\sum_{b_i \in B_1} b_i = \sum_{b_i \in B_2} b_i.$$

Put

$$\begin{aligned} e_1 &= 1, \\ e_{2i} &= e_{2i-1} + b_i, \quad i = 1, \dots, n_0, \\ e_{2i+1} &= e_{2i} + 1, \quad i = 1, \dots, n_0 - 1. \end{aligned}$$

Consider arbitrary subsets A_1 and A_2 containing by one element $a_i^1 \in A_1$ and $a_i^2 (\neq a_i^1) \in A_2$ from each pair a_{2i-1}, a_{2i} , $i = 1, \dots, n_0$. Then

$$\sum_{i=1}^{n_0-k} e_i^1 < \sum_{i=1}^{n_0} e_i^2 + \sum_{i=n_0-k+1}^{n_0} e_i^1, \quad k = 1, \dots, n_0 - 1,$$

for the constructed series of instances, since

$$\begin{aligned} \sum_{i=1}^{n_0-k} (e_i^1 - e_i^2) &\leq \sum_{i=1}^{n_0-k} b_i < \sum_{i=n_0-k+1}^{n_0} (e_i^1 + e_i^2) \\ &= \sum_{i=n_0-k+1}^{n_0} (2b_1 + 2b_2 + \dots + 2b_{i-1} + b_i + 2i), \\ &k = 1, \dots, n_0 - 1. \end{aligned}$$

So, only n_0 elements selected from pairs can give the total sum of weights equal to E .

Moreover

$$\begin{aligned} \sum_{a_i \in A_1} e_i &= \sum_{i=1}^{n_0} e_{2i-1} + \sum_{b_i \in B_1} b_i = \sum_{i=1}^{n_0} e_{2i-1} + \sum_{b_i \in B_2} b_i \\ &= \sum_{a_i \in A_2} e_i \end{aligned}$$

if and only if

$$\sum_{b_i \in B_1} b_i = \sum_{b_i \in B_2} b_i.$$

We will consider such NP-hard instances in our subsequent reductions. $\square$

Let $r_{\min} = \min_j r_j$ and $d_{\max} = \max_j d_j$ or $d_{\max}$ is equal to some other upper bound on the schedule length. We say that a schedule is called non-idle if the processor is no idle during the interval $[r_{\min}, d_{\max}]$. We say an instance of the problem has the non-idle property if there is no feasible schedule that is not non-idle. Our NP-hardness proofs are based on the polynomial reduction of the Ordered Partition Problem to the decision version of the scheduling problem with job prohibitions. The obtained instance of the decision problem has the non-idle property. Moreover, main jobs j correspond to elements a_j and job subsets X^i contain pairs a_{2i-1}, a_{2i} , but auxiliary critical jobs can be also introduced in some cases. At the same time, each reduction is unique due to the properties of the objective function and additional constraints.

Theorem 2 *Problem 1| $d_j = d$ | $\sum_j w_j U_j$ with job prohibitions is NP-hard even in the case of the system of subsets $X = \{X^i\}$, where $|X^i| = 2$, $i = 1, \dots, n$.*

Proof Let the number of jobs $n = 2n_0$. Put $p_j = w_j = e_j$ and $d_j = E$ for job $j \in \mathcal{J}$. Define job subsets as $X^{i+n_0} = X^i = \{2i-1, 2i\}$, $i = 1, \dots, n_0$.

Permutation π' satisfying prohibition constraints and having $\sum_j w_j U_j(\pi') \leq E$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 1). Indeed, the total due date E for all jobs guarantees that $\sum_j w_j U_j(\pi') \leq E$ can take place only as equality. Moreover, the subset J_1 of n_0 jobs (contains by one job from each pair $2i-1, 2i$) with the total duration E completed before the due date can exist only in the case when we have two subsets A_1, A_2 with sum of the weights equal to E in the Ordered Partition Problem (elements of A_1 correspond to jobs of J_1). $\square$

Theorem 3 *Problem 1| $r_j|\gamma$ with job prohibitions is NP-hard even in the case of the system of subsets $X = \{X^i\}$, such that $|X^i| \leq 2$, $i = 1, \dots, n$, for $\gamma \in \{L_{\max}; \sum_j U_j; \sum_j T_j\}$.*

Proof Set the number of jobs $n = 2n_0 + 1$. Let $p_j = e_j$, $d_j = 2E + 1$, $r_j = 0$ for jobs $j = 1, \dots, 2n_0$. Put $p_{2n_0+1} = 1$, $r_{2n_0+1} = E$ and deadline $d_{2n_0+1} = E + 1$ for job $2n_0 + 1$. Define the system X of job subsets as $X^{i+n_0+1} = X^i = \{2i-1, 2i\}$, $i = 1, \dots, n_0$ and $X^{n_0+1} = \{2n_0 + 1\}$.

Suggest that we search for a permutation π' satisfying the prohibition constraints and having objective $L_{\max}(\pi') \leq 0$ ($\sum_j U_j(\pi') \leq 0$ or $\sum_j T_j(\pi') \leq 0$). Job $2n_0 + 1$ enters in the system at time E , and must be completed at time $E + 1$. Moreover, the remaining jobs must be completed before time $2E + 1$. So, it is possible if and only if elements are partitioned into two parts A_1, A_2 of equal weights in the Ordered Partition Problem (see Fig. 2). $\square$

Theorem 4 *Problem 1| $C_j \leq d_j$ | $\sum_j w_j C_j$ with job prohibitions is NP-hard even in the case of the system of subsets $X = \{X^i\}$, where $|X^i| \leq 2$, $i = 1, \dots, n$, is NP-hard.*

Proof Suppose that the number of jobs $n = 2n_0 + 1$. Let $p_j = w_j = e_j$, $d_j = 2E + 1$ for jobs $j = 1, \dots, 2n_0$. For job $2n_0 + 1$ we put $p_{2n_0+1} = 1$, $w_{2n_0+1} = 0$ and $d_{2n_0+1} = E + 1$.

Define the system of job subsets as $X^{i+n_0+1} = X^i = \{2i-1, 2i\}$, $i = 1, \dots, n_0$ and $X^{n_0+1} = \{2n_0 + 1\}$.

Permutation π' satisfying prohibition constraints and having

$$\sum_j w_j C_j(\pi') \leq \sum_{1 \leq i \leq j \leq 2n_0} e_i e_j + E$$

exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 2). The weight is equal to the duration for a job $i = 1, \dots, 2n_0$. So, if job $2n_0 + 1$ is absent, then $\sum_{j=1}^{2n_0} w_j C_j = \sum_{1 \leq i \leq j \leq 2n_0} e_i e_j$ for any feasible ordering of the first $2n_0$ jobs. Job $2n_0 + 1$ of unit duration must be completed before time $E + 1$, therefore, the total weighted completion time $\sum_{j=1}^{2n_0+1} w_j C_j = \sum_{1 \leq i \leq j \leq 2n_0} e_i e_j + \sum_{j \in J_2} e_j$, where J_2 is the subset of jobs performed after job $2n_0 + 1$. As a result inequality $\sum_{j=1}^{2n_0+1} w_j C_j(\pi') \leq \sum_{1 \leq i \leq j \leq 2n_0} e_i e_j + E$ takes place if and only if subset J_2 contains jobs of the total weight E , and thus if and only if the Ordered Partition Problem has a positive answer with subsets A_1, A_2 . $\square$

Theorem 5 *Problem 1| r_j | $\sum_j C_j$ with job prohibitions is NP-hard even in the case of the system of subsets $X = \{X^i\}$, such that $|X^i| \leq 2$, $i = 1, \dots, n$, is NP-hard.*

Proof Suppose that we have $n = 2n_0 + 2$ jobs. Let $p_j = e_j$, $r_j = 0$ for jobs $j = 1, \dots, 2n_0$, and $p_{2n_0+1} = p_{2n_0+2} = 1$, $r_{2n_0+1} = E$, $r_{2n_0+2} = 2E + 1$ for jobs $2n_0 + 1, 2n_0 + 2$. Put subsets of jobs in the system X as $X^{i+n_0+1} = X^i = \{2i-1, 2i\}$, $i = 1, \dots, n_0$ and $X^{n_0+1} = \{2n_0 + 1\}$, $X^{2n_0+2} = \{2n_0 + 2\}$.

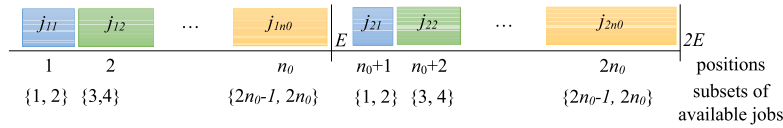


Fig. 1 Illustrative example to the reduction of the Ordered Partition Problem to $1|d_j = d|\sum_j w_j U_j$ with job prohibitions. Here we present a scheme of a non-idle schedule in which jobs form $\{j_{11}, \dots, j_{1n_0}\}$ complete before time E , jobs form $\{j_{21}, \dots, j_{2n_0}\}$ start after time E and

prohibition constraints are satisfied. Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form have objective $\sum_j w_j U_j \leq E$, all other schedules have greater objective values

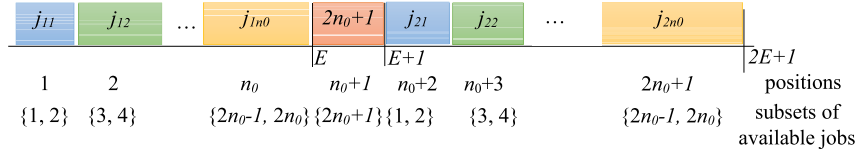


Fig. 2 Illustrative example to the reduction of the Ordered Partition Problem to $1|r_j|\gamma, \gamma \in \{L_{\max}, \sum_j U_j; \sum_j T_j\}$ and $1|C_j \leq d_j|\sum_j w_j C_j$ with job prohibitions. Here we present a scheme of a non-idle schedule, where the $(n_0 + 1)$ -th job is executed in interval $[E, E + 1)$, jobs from $\{j_{11}, \dots, j_{1n_0}\}$ complete before time E , jobs

form $\{j_{21}, \dots, j_{2n_0}\}$ start after time $E + 1$ and complete before time $2E + 1$, and all jobs are placed according to the system of job subsets X . These schedules exist if and only if the corresponding Ordered Partition Instance has a positive answer. Only such schedules have the objective value less than the given upper bound

Show that there is a permutation π' satisfying prohibition constraints and $\sum_j C_j(\pi') \leq L := (E + 1) + (2E + 2) + \left(\sum_{j=1}^{n_0} (n_0 - j + 1)(e_{2j-1} + e_{2j}) + (E + 1)n_0\right)$ if and only if the Ordered Partition Problem has a positive answer (see Fig. 3).

Let the Ordered Partition Problem have a positive answer, and A_1 and A_2 be subsets of A such that $\sum_{a_i \in A_1} e_i = \sum_{a_i \in A_2} e_i = E$, $|A_1| = |A_2| = n_0$. Denote by $\pi(A_1)$ and $\pi(A_2)$ permutations of jobs corresponding to elements form A_1 and A_2 , where jobs are sequenced in the non-decreasing order of their numbers. Then we have $\sum_j C_j(\pi') = L$ for $\pi' = (\pi(A_1), 2n_0 + 1, \pi(A_2), 2n_0 + 2)$.

Let π' be the permutation satisfying prohibition constraints and $\sum_j C_j(\pi') \leq L$. In this permutation job $2n_0 + 1$ is performed in position $n_0 + 1$ and the last position contains job $2n_0 + 2$. Denote by J_1 the subset of jobs performed before job $2n_0 + 1$ and by J_2 the subset of jobs executed after job $2n_0 + 1$ and before job $2n_0 + 2$ ($|J_1| = |J_2| = n_0$). It is required to prove that the total duration of jobs from J_1 is equal to E . Suppose that $\sum_{j \in J_1} p_j < E$, then $\sum_{j \in J_2} p_j > E$ and $\sum_j C_j(\pi') \geq L + 1$ as job $2n_0 + 2$ is completed at time $2E + 3$ or later. If the total duration of jobs from J_1 is greater than E , then $\sum_j C_j(\pi') \geq L + 1 + (E + 2)n_0 - (E + 1)n_0 = L + 1 + n_0$ since jobs from J_2 start execution after time moment $(E + 2)$, and job $2n_0 + 1$ starts not early than time moment $(E + 1)$. Therefore, $\sum_{j \in J_1} p_j = \sum_{j \in J_2} p_j = E$ and subsets $A_1 = \{a_j : j \in J_1\}$, $A_2 = \{a_j : j \in J_2\}$ satisfy the conditions of the Ordered Partition Problem.

□

Theorem 6 Problems $2||C_{\max}, 2|d_j = d|\gamma, \gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions are NP-hard for the systems of job subsets $X = \{X^{i,l}\}$, $|X^{i,l}| = 2$, $i \in N_l$, $l = 1, 2$.

Proof Consider the individual scheduling problem with $n = 2n_0$ jobs. Let durations $p_j = e_j$ for jobs $j = 1, \dots, 2n_0$.

Now we define system X of job subsets as $X^{i,1} = X^{i,2} = \{2i - 1, 2i\}$, $i = 1, \dots, n_0$ and $X^{i,1} = X^{i,2} = \emptyset$, $i = 1 + n_0, \dots, 2n_0$.

It is easy to see that family of permutations $\{\pi^l\}_{l \in M}$ satisfying prohibition constraints and $C_{\max} \leq E$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 4). It follows from the property that all jobs should be finished before time moment E . It is possible only in the case when elements from A can be divided into two parts A_1 , A_2 with equal total weight in the Ordered Partition Problem. For criteria $L_{\max}, \sum_j U_j, \sum_j T_j$ we use the same reduction scheme, additionally setting $r_j = 0$, $d_j = E$ and threshold equal to zero.

□

Theorem 7 Ereemeev and Kovalenko (2014) Problem $1||l_{ij}|C_{\max}$ with job prohibitions is NP-hard for the systems of job subsets $X = \{X^i\}$, $|X^i| \leq 2$, $i = 1, \dots, n$.

Theorem 8 Zakharova (2023a) Problem $1||\sum_j w_j T_j$ with job prohibitions is NP-hard for the systems of job subsets $X = \{X^i\}$, $|X^i| \leq 2$, $i = 1, \dots, n$, is NP-hard.

NP-hardness of the optimal recombination problems was proved in papers (Ereemeev & Kovalenko, 2014; Zakharova,

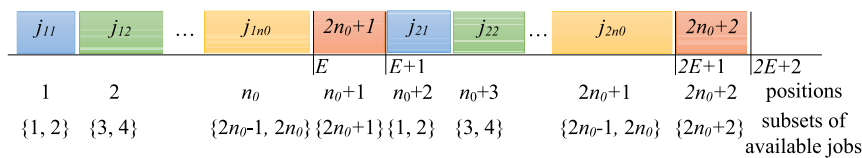


Fig. 3 Illustrative example to the reduction of the Ordered Partition Problem to $1|r_j|\sum_j C_j$ with job prohibitions. Here we present a scheme of a non-idle schedule, where job $2n_0 + 1$ is executed in interval $[E, E + 1)$, job $2n_0 + 2$ is performed in interval $[2E + 1, 2E + 2)$ and the rest jobs are partitioned into two parts $J_1 = \{j_{11}, \dots, j_{1n_0}\}$,

$J_2 = \{j_{21}, \dots, j_{2n_0}\}$ satisfying prohibition constraints and having the total duration equal to E . Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form have objective $\sum_j C_j \leq L$, all other schedules have greater objective values

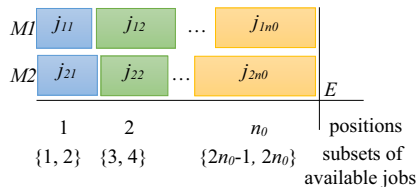


Fig. 4 Illustrative example to the reduction of the Ordered Partition Problem to $2||C_{\max}, 2|d_j = d|\gamma, \gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions. Here we present a scheme of a non-idle schedule in which the last job on both machines completes at time E and prohibition constraints are satisfied. Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form (subsets of jobs $J_1 = \{j_{11}, \dots, j_{1n_0}\}$ and $J_2 = \{j_{21}, \dots, j_{2n_0}\}$ performed on machines $M1$ and $M2$) have objective value less than the threshold

2023a), but NP-hardness of the corresponding scheduling problems with job prohibitions is proved in the similar way. In the first case we use a polynomial reduction from the special series of the problem of searching the maximum clique of the minimal weight as in the paper of Serdyukov (1978). In the second case the Restricted Even-Odd Partition Problem (Du & Leung, 1990) is reduced for proving NP-hardness.

2.2 Solving approach

Consider an individual instance $I(\gamma, X^i)$ of a single processor scheduling problem with objective function $\gamma(\cdot)$ and a system of job subsets $X = \{X^i\}_{i=1, \dots, n}$, where $|X^i| \leq 2$. We will use the same exact approach as proposed in papers (Eremeev & Kovalenko, 2017; Serdyukov, 1978). Now we construct a bipartite graph $G = (X_n, X, U)$ with two parts of the vertices set X_n, X of equal sizes and the set of edges $U = \{(i, x) : i \in X_n, x \in X^i\}$. The left part of the vertices set represent positions and the right part shows jobs. By construction we see that there is a one-to-one correspondence between perfect matchings of G and feasible permutations of instance $I(\gamma, X^i)$. Recall that the set of edges of a perfect matching “covers” all vertices of the graph. We will denote the set of perfect matchings by $\mathcal{W}$, and the set of feasible permutations to $I(\gamma, X^i)$ by Π .

According to terminology from Eremeev and Kovalenko (2017); Serdyukov (1978) we have the following types of edges in graph G . *Special* edges belong to all perfect matchings in G , and edges adjacent to them are absent in all perfect matching of G . The last ones can be found by Algorithm $\mathcal{A}$ from Eremeev and Kovalenko (2017).

Algorithm $\mathcal{A}$ (Eremeev & Kovalenko, 2017).

1. Put auxiliary graph $G' := G$, set of special edges $U' := \emptyset$.
2. Repeat until the termination condition is satisfied or G' is 2-regular:
 - 2.1. If G' contains a vertex z of degree 0, then the problem is infeasible, stop.
 - 2.2. If G' contains a vertex z of degree 1, then $U' := U' \cup \{z, y\}$ and remove y, z from G' .
3. $G' := G' \cup U'$.

The total number of edges in G and G' is bounded by $2n$, and each edge is visited only once, therefore, the running time of Algorithm $\mathcal{A}$ is $O(n)$. In addition, we will identify by $\mathcal{A}$ the case when the problem is infeasible. So, we will investigate only feasible instances and put $G := G'$.

The non-special edges of G are slit into even cycles (see Fig. 5), as proven in Eremeev and Kovalenko (2017). The Depth-First Search algorithm (Cormen et al., 2001) can be used for finding cycles of G in $O(n)$ time. Let $q(G) = q(I)$ denote the number of cycles in graph G generated for instance $I(\gamma, X^i)$.

A cycle $j, j = 1, \dots, q(G)$, has two perfect matchings. And a perfect matching in G is composed by perfect matchings of cycles and special edges (see Fig. 5), as shown in Eremeev and Kovalenko (2017).

The proposed algorithm is based on the same principles as other algorithms for enumerating perfect matchings in bipartite graphs (see, e.g., Uno, 1997), but it has less running time due to the property that vertices of the left part have degree at most two.

So, the scheduling problem with job prohibitions $I(\gamma, X^i)$ can be solved as follows. We enumerate all perfect matchings $W \in \mathcal{W}$, constructing the corresponding permutation $\pi \in \Pi$ to each $W \in \mathcal{W}$ and compute $\gamma(\pi)$. The permutation with the optimum objective value gives us the optimal solution. Note that $|\Pi| = |\mathcal{W}| = 2^{q(G)} = 2^{q(I)}$.

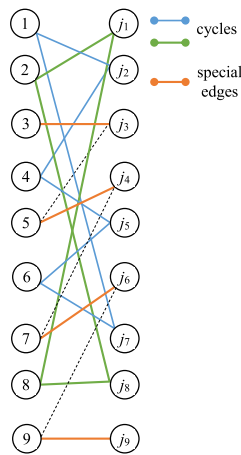


Fig. 5 An instance with $n = 9$ jobs and the system X of job subsets: $X^1 = \{j_2, j_7\}$, $X^2 = \{j_1, j_8\}$, $X^3 = \{j_3\}$, $X^4 = \{j_2, j_5\}$, $X^5 = \{j_3, j_4\}$, $X^6 = \{j_5, j_7\}$, $X^7 = \{j_4, j_6\}$, $X^8 = \{j_1, j_8\}$, $X^9 = \{j_6, j_9\}$. The corresponding bipartite graph $G = (X_9, X, U)$ has the presented form. We see two cycles (blue and green colors) and four special edges (red color). Four feasible solutions are $(j_2, j_1, j_3, j_5, j_4, j_7, j_6, j_8, j_9)$, $(j_2, j_8, j_3, j_5, j_4, j_7, j_6, j_1, j_9)$, $(j_7, j_1, j_3, j_2, j_4, j_5, j_6, j_8, j_9)$, $(j_7, j_8, j_3, j_2, j_4, j_5, j_6, j_1, j_9)$ (Color figure online)

The provided algorithm has running time $O(n + T(\gamma)2^{q(I)})$, where $T(\gamma)$ is the running time of computing objective γ , $q(I) \leq \lfloor \frac{n}{2} \rfloor$. Thus, similar to the previous results for the Traveling Salesman Problem with prohibitions (Eremeev & Kovalenko, 2017; Serdyukov, 1978), we obtain

Theorem 9 *Scheduling problems $1|r_j, d_j|\gamma$, $\gamma \in \{C_{\max}, L_{\max}, \sum_j w_j C_j, \sum_j w_j T_j, \sum_j w_j U_j\}$, with job prohibitions can be solved in $O(n1.42^n)$ time for a system of job subsets $X = \{X^i\}_{i=1, \dots, n}$ such that $|X^i| \leq 2$, $i = 1, \dots, n$.*

According to the terminology from Eremeev and Kovalenko (2017), Serdyukov (1978), a bipartite graph $G = (X_n, X, U)$ is called “good” if the number of cycles $q(G)$ is bounded by $1.1 \ln n$. Denote by $\bar{\chi}_n$ the set of “good” bipartite graphs $G = (X_n, X, U)$, and by χ_n the set of all bipartite graphs $G = (X_n, X, U)$.

Theorem 10 $\frac{|\bar{\chi}_n|}{|\chi_n|} \rightarrow 1$ as $n \rightarrow \infty$ (Eremeev & Kovalenko, 2014; Serdyukov, 1978).

As a result we conclude that almost all systems of job subsets with $|X^i| \leq 2$, $i = 1, \dots, n$, provide instances $I(\gamma, X^i)$ having at most n feasible solutions and thus, they are solvable in $O(n^2)$ time.

Note that for an arbitrary system of job subsets (without bounds on the cardinality) a feasible schedule can be found in polynomial time by solving the well-known assignment problem (Munkres, 1957), and the optimal solution may be computed in $O(n^{2.5} + n^2 n_{pm})$ time by enumerating perfect matchings (see, e.g., Uno, 1997) in the corresponding

bipartite graph (n_{pm} is the number of perfect matchings) and computing objectives for appropriate schedules. Here time $O(n^{2.5})$ corresponds to compute one perfect matching and time $O(n^2 n_{pm})$ is used for going from the current perfect matching to the next one and calculating the required metrics.

2.3 Integer programming models

Now we consider an adaptation of the classic model with position-based solution representation (Blazewicz et al., 1991) to the single machine problem under job prohibition constraints.

Consider the following main variables:

$$x_{ij} = \begin{cases} 1, & \text{if job } j \text{ is performed in position } i, \\ 0 & \text{otherwise, } i = 1, \dots, n, j \in X^i; \end{cases}$$

$C_i \geq 0$ is the completion time of the job in position i .

Introduce auxiliary variables for simplifying the model description:

$y_i \geq 0$ is the duration of the job in position i ;

$z_i \geq 0$ is the release date of the job in position i ;

$v_i \geq 0$ is the due date of the job in position i .

Denote by Y^j the subset of positions, where job $j \in \mathcal{J}$ can be performed, i.e. $Y^j = \{i \in X_n : j \in X^i\}$.

Our model has the following constraints:

$$\sum_{j \in X^i} x_{ij} = 1, i = 1, \dots, n, \quad (1)$$

$$\sum_{i \in Y^j} x_{ij} = 1, j \in \mathcal{J}, \quad (2)$$

$$y_i = \sum_{j \in X^i} x_{ij} p_j, i = 1, \dots, n, \quad (3)$$

$$z_i = \sum_{j \in X^i} x_{ij} r_j, i = 1, \dots, n, \quad (4)$$

$$v_i = \sum_{j \in X^i} x_{ij} d_j, i = 1, \dots, n, \quad (5)$$

$$C_i \geq C_{i-1} + y_i, i = 2, \dots, n, \quad (6)$$

$$C_i \geq z_i + y_i, i = 1, \dots, n, \quad (7)$$

$$C_1 \geq 0, \quad (8)$$

$$x_{ij} \in \{0, 1\}, i = 1, \dots, n, j \in X^i. \quad (9)$$

Constraints (1)–(2) ensure that a position contains exactly one job, and each job is executed in one position. Equality (3)–(5) compute durations, release dates and due dates of jobs in positions (auxiliary variables). Constraint (6) indicates that the completion time in position i should be not less than the completion time in the preceding position plus the duration of the job in position i . Condition (7) takes into account release

dates of jobs in computing completion times. Constraints (8)–(9) give the area where the variables are defined. Setup times are modeled in the same manner as, for example, in paper (Eremeev & Kovalenko, 2011).

Criteria can be written as:
the maximum lateness

$$L_{\max} \rightarrow \min,$$

$$L_{\max} \geq C_i - v_i, \quad i = 1, \dots, n,$$

the total tardiness

$$\sum T_i = \sum_{i=1}^n T_i \rightarrow \min,$$

$$T_i \geq 0, \quad T_i \geq C_i - v_i, \quad i = 1, \dots, n,$$

the total completion time

$$\sum C_i = \sum_{i=1}^n C_i \rightarrow \min,$$

the number of tardy jobs

$$\sum U_i = \sum_{i=1}^n U_i \rightarrow \min,$$

$$C_i \leq v_i + U_i \cdot M, \quad U_i \in \{0, 1\}, \quad i = 1, \dots, n.$$

here and later we denote by M a sufficiently big constant.

So, this model has $n \leq \sum_{i=1}^n |X^i| \leq 2n$ Boolean variables and n continuous variables for the case $|X^i| \leq 2$, $i = 1, \dots, n$. Note that the model does not use additional information about relations of the solutions for the individual problem instance to the perfect matchings of the special bipartite graph, and can be also used for the arbitrary system of job subsets $X = \{X^i\}_{i=1, \dots, n}$. Adjacency-based models are also known in the literature for scheduling problems (see, e.g. Blazewicz et al., 1991), but such models are less appropriate for prohibition constraints.

Now we present a new idea for formulating an integer programming model for single-machine scheduling problems with a system of job subsets X^i for positions $i = 1, \dots, n$ such that $|X^i| \leq 2$. This model uses preliminary computations of special edges, cycles, and maximal matchings of cycles in the corresponding bipartite graph. According to the results of Sect. 2.2 we will consider only feasible instances and can assume that all positions correspond to special edges or cycles.

Let us introduce the following notations:

N_l is the set of positions corresponding to cycle l , $l = 1, \dots, q(G)$;

l_i is the cycle, containing position i (if position i correspond to a special edge, then we put $l_i = 0$);

p_i^0 (p_i^1) is the duration of the job in position i , that corresponds to the first (the second) maximal matching of cycle l_i (if $l_i = 0$, then $p_i^0 = p_i^1$);

r_i^0 (r_i^1) is the release date of the job in position i , that corresponds to the first (the second) maximal matching of cycle l_i (if $l_i = 0$, then $r_i^0 = r_i^1$);

d_i^0 (d_i^1) is the due date of the job in position i , that corresponds to the first (the second) maximal matching of cycle l_i (if $l_i = 0$, then $d_i^0 = d_i^1$);

Introduce the following Boolean variables:

$$x_l = \begin{cases} 0, & \text{if the first matching is selected in cycle } l, \\ 1, & \text{if the second matching is selected in cycle } l; \end{cases}$$

Let variable C_i and auxiliary variables y_i , z_i and v_i have the same sense as in the previous model for positions $i = 1, \dots, n$. Note that Boolean variables of such type are also used in a mathematical programming model for the Traveling Salesman Problem with prohibitions (Eremeev & Kovalenko, 2017), but here we use properties of scheduling problems and their metrics.

Then the mixed integer linear programming model has the following form:

$$y_i = p_i^0(1 - x_l) + p_i^1 x_l, \quad l = 1, \dots, q(G), \quad i \in N_l, \quad (10)$$

$$y_i = p_i^0, \quad i = 1, \dots, n : |X^i| = 1,$$

$$z_i = r_i^0(1 - x_l) + r_i^1 x_l, \quad l = 1, \dots, q(G), \quad i \in N_l, \quad (11)$$

$$z_i = r_i^0, \quad i = 1, \dots, n : |X^i| = 1,$$

$$v_i = d_i^0(1 - x_l) + d_i^1 x_l, \quad l = 1, \dots, q(G), \quad i \in N_l, \quad (12)$$

$$v_i = d_i^0, \quad i = 1, \dots, n : |X^i| = 1,$$

$$C_i \geq C_{i-1} + y_i, \quad i = 2, \dots, n, \quad (13)$$

$$C_i \geq z_i + y_i, \quad i = 1, \dots, n, \quad (14)$$

$$C_1 \geq 0, \quad (15)$$

$$x_l \in \{0, 1\}, \quad l = 1, \dots, q(G). \quad (16)$$

Equality (10)–(12) identify auxiliary variables: durations, release dates and due dates of jobs in positions. Constraints (13) and (14) compute completion times in positions taking into account the previous positions and release dates of jobs. Constraints (15)–(16) give the feasible area of variables. Here setup times can be modeled as in Eremeev and Kovalenko (2017). The approach is based on the contacts between cycles, corresponding to consecutive positions of a permutation.

Criteria are written as in the previous model. The number of real variables in the model is n , the number of Boolean variables is $q(G) \leq \lfloor \frac{n}{2} \rfloor$. The number of constraints is $O(n)$. The proposed model may be used for computing lower bound of the objective function, or in branch-and-bound algorithms.

machines	$M1$	$\{j_1, j_5\}$	$\{j_9, j_{10}\}$		
	$M2$	$\{j_4, j_7\}$	$\{j_2, j_8\}$	$\{j_6, j_8\}$	$\{j_2, j_6\}$
	$M3$	$\{j_1, j_9\}$	$\{j_5, j_{10}\}$	$\{j_3, j_7\}$	
	$M4$	$\{j_3, j_4\}$	$\{j_{11}\}$		
	positions	1	2	3	4

Fig. 6 Illustrative example to the multi-machine case. Here we present input data for the instance with four machines and eleven jobs. Job indicators in braces represent subsets of available jobs in positions of machines. We have 8 solutions (presented in accordance with the common sequence of positions): $(j_1, j_{10}, j_4, j_2, j_8, j_6, j_9, j_5, j_7, j_3, j_{11})$, $(j_1, j_{10}, j_7, j_2, j_8, j_6, j_9, j_5, j_3, j_4, j_{11})$, $(j_1, j_{10}, j_4, j_8, j_6, j_2, j_9, j_5, j_7, j_3, j_{11})$, $(j_1, j_{10}, j_7, j_8, j_6, j_2, j_9, j_5, j_3, j_4, j_{11})$, $(j_5, j_9, j_4, j_2, j_8, j_6, j_1, j_{10}, j_7, j_3, j_{11})$, $(j_5, j_9, j_7, j_2, j_8, j_6, j_1, j_{10}, j_3, j_4, j_{11})$, $(j_5, j_9, j_4, j_8, j_6, j_2, j_1, j_{10}, j_7, j_3, j_{11})$, $(j_5, j_9, j_7, j_8, j_6, j_2, j_1, j_{10}, j_3, j_4, j_{11})$

2.4 Multi-machine system

Now we consider the system with $m > 1$ machines. Suppose that the total number of positions over all machines is equal to the number of jobs, and all jobs are independent (see an example in Fig. 6). Then the multi-machine problem can be solved using the same methods as the single-machine one. Here we consider the common sequence of positions $(1, \dots, n)$ for all machines, where i is the i' -th position on machine l' defined as $i' = i - \sum_{l=1}^{l'-1} |N_l| \leq |N_{l'}|$. All theorems and models from Sects. 2.1, 2.2, 2.3 take place for such multi-machine case.

In the case, when subsets of jobs of different machines do not intersect, the problem is decomposed into m independent subproblems.

3 Multistage system

In this section we consider the multistage systems (job-shop, flow-shop and open-shop) with job prohibitions. The computational complexity is analyzed and solving methods are provided.

3.1 Job-shop and flow-shop scheduling with job prohibitions

The job-shop scheduling problem with job prohibitions is defined as follows: a set $\mathcal{J} = \{1, \dots, n\}$ of n jobs is to be processed in the system with a set $\mathcal{M} = \{1, \dots, m\}$ of m machines. A job j consists of k_j sequential operations. Operation O_v^j of job j with number v , $v = 1, \dots, k_j$, has duration p_{vj} and must be performed on machine $L_{vj} \in \mathcal{M}$. Preemptions of operations are disallowed. At any time each

machine can process at most one job, and each job can be processed on at most one machine.

Sequences of machines $L_j = (L_{1j}, L_{2j}, \dots, L_{k_j, j})$, $j \in \mathcal{J}$, may be different for different jobs. Let $O_l := \{O_v^j : L_{vj} = l\}$, $|O_l| = n_l$, denote the set of operations, that must be performed on machine $l \in \mathcal{M}$. Available jobs are given by subset X^{il} for position $i = 1, \dots, n_l$ of machine $l \in \mathcal{M}$.

We consider the objective functions to be minimized: the makespan, $C_{\max} = \max\{C_j : j \in \mathcal{J}\}$, the total completion time, $\sum_j C_j$, the maximum lateness, $L_{\max} = \max\{C_j - d_j : j \in \mathcal{J}\}$, the total tardiness, $\sum_j T_j = \sum_{j \in \mathcal{J}} \max\{C_j - d_j, 0\}$, the number of tardy jobs, $\sum_j U_j = \sum_{j \in \mathcal{J}} \chi_{C_j > d_j}$, where d_j is the due date for job j and C_j is the completion time of job j , $j \in \mathcal{J}$.

Let $\pi^l = (\pi_1^l, \dots, \pi_{n_l}^l)$ denote a permutation of operations on machine l , i.e. π_i^l is the i -th job on machine $l \in \mathcal{M}$, $i = 1, \dots, n_l$. For a given family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$, the early schedule and the corresponding value of objective function $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}})$, $L_{\max}(\{\pi^l\}_{l \in \mathcal{M}})$, $\sum_j C_j(\{\pi^l\}_{l \in \mathcal{M}})$, $\sum_j T_j(\{\pi^l\}_{l \in \mathcal{M}})$, $\sum_j U_j(\{\pi^l\}_{l \in \mathcal{M}})$ are calculated as follows.

Construct the weighted directed acyclic graph $G = (V, A)$ with the set of vertices V and the set of arcs A , where $|V| = \sum_{j \in \mathcal{J}} k_j$ and vertices correspond to operations. The set of arcs A defines precedence constraints on the set of operations, given by the family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$ and the sequences of machines for jobs L_j , $j \in \mathcal{J}$. The weight (duration) of operation-vertex O_v^j is set to p_{vj} . Then the early schedule for the constructed graph may be computed by the algorithm known as the topological sort (Cormen et al., 2001) in time

$$\begin{aligned} O(|V| + |A|) &= O\left(\sum_{j \in \mathcal{J}} k_j + \left(\sum_{j \in \mathcal{J}} (k_j - 1) + \sum_{l \in \mathcal{M}} (n_l - 1)\right)\right) \\ &= O(nk_{\max} + mn_{\max}), \end{aligned}$$

where $k_{\max} = \max_{j \in \mathcal{J}} k_j$ and $n_{\max} = \max_{l \in \mathcal{M}} n_l$.

Later on, we suggest that the schedule for family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$ corresponds to the early schedule for this family, and objective value is represented by the objective value for the early schedule.

Let $C_{vj}(\{\pi^l\}_{l \in \mathcal{M}})$ is the completion time of operation O_v^j of job j in the early schedule for family $\{\pi^l\}_{l \in \mathcal{M}}$, then the makespan

$$C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) = \max \left\{ C_{k_j, j}(\{\pi^l\}_{l \in \mathcal{M}}) : j \in \mathcal{J} \right\},$$

the total completion time

$$\sum_j C_j(\{\pi^l\}_{l \in \mathcal{M}}) = \sum_{j \in \mathcal{J}} C_{k_j, j}(\{\pi^l\}_{l \in \mathcal{M}}),$$

the maximum lateness

$$L_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) = \{C_{k_j, j}(\{\pi^l\}_{l \in \mathcal{M}}) - d_j : j \in \mathcal{J}\},$$

the total tardiness

$$\sum_j T_j(\{\pi^l\}_{l \in \mathcal{M}}) = \sum_{j \in \mathcal{J}} \max \{C_{k_j, j}(\{\pi^l\}_{l \in \mathcal{M}}) - d_j; 0\},$$

the number of tardy jobs

$$\sum_j U_j(\{\pi^l\}_{l \in \mathcal{M}}) = \sum_{j \in \mathcal{J}} \chi_{C_{k_j, j}(\{\pi^l\}_{l \in \mathcal{M}}) > d_j}.$$

Job-shop scheduling problem $J||\gamma$, $\gamma \in \{C_{\max}, L_{\max}, \sum_j C_j, \sum_j T_j, \sum_j U_j\}$ with job prohibitions is formulated as the problem of finding a family of permutations $\{\pi^{*l}\}_{l \in \mathcal{M}}$ that minimizes the objective function γ and places jobs from the given subsets X^{il} in positions $i = 1, \dots, n_l$ of machines $l \in \mathcal{M}$.

The flow-shop scheduling problem is the particular case of the job-shop when all jobs have identical routes (sequences of operations), i.e. $L_j = (1, 2, \dots, m)$ for $j \in \mathcal{J}$. In the classic case of flow-shop $p_{vj} = 0$ means that job j is executed on machine v , but the duration of its service can be neglected (denoted by $F||\gamma$ (Graham et al., 1979)). We will investigate the classic case and also the case, when $p_{vj} = 0$ means that job j on machine v is not executed. Such problem is called the flow-shop scheduling with missing operations of jobs and denoted by $F|Miss - Oper||\gamma$ (Graham et al., 1979). It is the job-shop problem of flow-shop type. We will also investigate flow-shop problems with job prohibitions. In this case the early schedule and objectives are computed in $O(nm)$ time using the topological sort (Cormen et al., 2001).

Firstly, we prove NP-hardness of the two-machine job-shop with different routes of jobs and prohibitions. Secondly, we investigate the computational complexity of the flow-shop with prohibitions, when all jobs have identical routes.

Theorem 11 *Job-shop scheduling problem $J2||C_{\max}$ with job prohibitions is NP-hard even for the system of subsets $X = \{X^{i,l}\}$, such that $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

Proof Put $m = 2$ and $n = 2n_0 + 2$. Let durations

$$\begin{aligned} p_{1,j} &= 0, \quad p_{2,j} = e_j, \quad j = 1, \dots, 2n_0; \\ p_{1,2n_0+1} &= 2E, \quad p_{2,2n_0+1} = E; \\ p_{2,2n_0+2} &= 2E, \quad p_{1,2n_0+2} = E. \end{aligned}$$

Jobs $j = 1, \dots, 2n_0$ consist of one operation, that must be executed on the second machine. Job $2n_0 + 1$ consists of two operations, the first one is performed on the first machine and the second one requires the second machine. Job $2n_0 + 2$

also consists of two operations, but the first one is performed on the second machine and the second one requires the first machine. So, these two jobs have different routes.

We define the system of job subsets as

$$X^{11} = \{2n_0 + 1\}, \quad X^{21} = \{2n_0 + 2\}, \quad (17)$$

$$X^{i+n_0+2,2} = X^{i,2} = \{2i - 1, 2i\}, \quad i = 1, \dots, n_0, \quad (18)$$

$$X^{n_0+1,2} = \{2n_0 + 2\}, \quad X^{n_0+2,2} = \{2n_0 + 1\}. \quad (19)$$

Now we show that family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$ taking into account job prohibitions and having $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 7).

- (1) Suggest that the Ordered Partition Problem has a positive answer and subsets A_1, A_2 of set A are such that $\sum_{a_i \in A_1} e_i = \sum_{a_i \in A_2} e_i = E$ and $|A_1| = |A_2| = n_0$. We define permutations of jobs $J_1 = (j'_1, \dots, j'_{n_0})$ and $J_2 = (j''_1, \dots, j''_{n_0})$, corresponding to elements from A_1 and A_2 , respectively. Here $j'_i < j'_{i+1}$ and $j''_i < j''_{i+1}$ for all $i = 1, \dots, n_0 - 1$. Then permutations $\{\pi^1 = (2n_0 + 1, 2n_0 + 2), \pi^2 = (J_1, 2n_0 + 2, 2n_0 + 1, J_2)\}$ satisfy conditions (17)–(19) and $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) = 4E$. Both machines work without idle time in interval $(0, 4E]$ in the corresponding schedule.
- (2) Consider a family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$, that satisfies conditions (17)–(19) and has $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$. From the feasibility of the schedule we state that job $2n_0 + 1$ starts at time 0, job $2n_0 + 2$ starts at time $2E$ on the first machine and $C_{1,2n_0+1} = 2E$, $C_{2,2n_0+2} = 4E$. Therefore, job $2n_0 + 1$ is completed at time $t' \geq 3E$ and job $2n_0 + 2$ is completed at time $t'' \leq 2E$ on the second machine.

Let J_1 be the subset of jobs, presented in permutation π^2 before job $2n_0 + 2$, and J_2 be the subset of jobs, placed in π^2 after job $2n_0 + 1$. Then subsets $A_1 = \{a_i \in A : i \in J_1\}$ and $A_2 = \{a_i \in A : i \in J_2\}$ satisfy conditions of the Ordered Partition Problem.

The presented transformation is polynomial in n_0 . So, problem $J2||C_{\max}$ with job prohibitions is NP-hard. $\square$

Note that the problem $J2||C_{\max}$ without job prohibitions is polynomially solvable, when each job consists of at most two operations. It becomes to be NP-hard in the case of three machines (Tanaev et al., 1989).

If we have $d_j = 4E$, $j \in \mathcal{J}$, in problem $J||\gamma$, $\gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, then Theorem 11 guarantees that

Theorem 12 *Job-shop scheduling problem $J2|d_j = d||\gamma$, $\gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions is NP-hard*

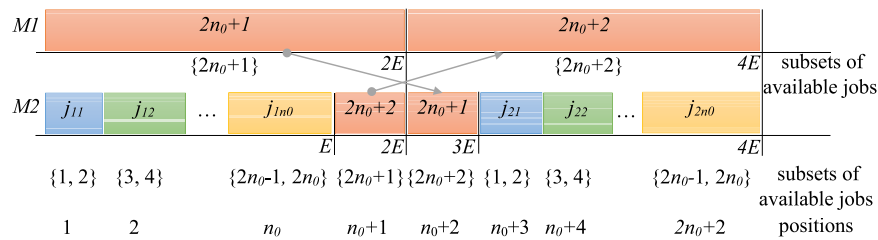


Fig. 7 Illustrative example to the reduction of the Ordered Partition Problem to $J2||C_{\max}$ with job prohibitions. The arrows indicate precedence relations between jobs. Here we present a scheme of a non-idle schedule, where job $(2n_0 + 1)$ is executed in interval $[0, 3E)$, job $(2n_0 + 2)$ is performed in interval $[E, 4E)$ and the rest jobs are divided into two parts $J_1 = \{j_{11}, \dots, j_{1n_0}\}$ and $J_2 = \{j_{21}, \dots, j_{2n_0}\}$ in accor-

dance with the system of available jobs in positions of machines. Both parts have the same total duration E . Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form have objective $C_{\max} \leq 4E$. All other schedules have greater objective values

even for the system of job subsets $X = \{X^{il}\}$, where $|X^{il}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.

Theorem 13 *Flow-shop scheduling problem $F3|Miss - Oper|C_{\max}$ with job prohibitions is NP-hard even for the system of job subsets X^{il} , where $|X^{il}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$, and each job consists of two operations with nonzero duration.*

Proof We construct a series of instances for $F|Miss - Oper|C_{\max}$ with $m = 3$ and $n = 2n_0 + 2$. Put processing times of jobs

$$\begin{aligned} p_{1,i} &= p_{3,i} = e_i, \quad p_{2,i} = 0, \quad i = 1, \dots, 2n_0; \\ p_{1,2n_0+1} &= p_{3,2n_0+2} = E, \quad p_{2,2n_0+1} = p_{2,2n_0+2} = 2E, \\ p_{3,2n_0+1} &= p_{1,2n_0+2} = 0. \end{aligned}$$

Suggest that the system of job subsets has the following structure

$$X^{i,1} = X^{n_0+1+i,1} = \{2i - 1, 2i\}, \quad i = 1, \dots, n_0, \quad (20)$$

$$\begin{aligned} X^{n_0+1,1} &= \{2n_0 + 1\}, \\ X^{1,2} &= \{2n_0 + 2\}, \quad X^{2,2} = \{2n_0 + 1\}, \end{aligned} \quad (21)$$

$$\begin{aligned} X^{i,3} &= X^{n_0+1+i,3} = \{2i - 1, 2i\}, \quad i = 1, \dots, n_0, \\ X^{n_0+1,3} &= \{2n_0 + 2\}. \end{aligned} \quad (22)$$

We will show that family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$ taking into account job prohibitions and $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 8).

- (1) Let the Ordered Partition Problem has a positive answer and subsets A_1, A_2 of set A are such that $\sum_{a_i \in A_1} e_i = \sum_{a_i \in A_2} e_i = E$ and $|A_1| = |A_2| = n_0$. Denote by $N_1 = (j_1, \dots, j_{n_0})$ and $N_2 = (j'_1, \dots, j'_{n_0})$ the permutations of jobs, corresponding to elements of A_1

and A_2 , respectively, where $j_i < j_{i+1}$ and $j'_i < j'_{i+1}$ for all $i = 1, \dots, n_0 - 1$. Then job prohibitions and threshold constraint $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) = 4E$ take place for the family of permutations: $\{\pi^1 = (N_1, 2n_0 + 1, N_2), \pi^2 = (2n_0 + 2, 2n_0 + 1), \pi^3 = (N_1, 2n_0 + 2, N_2(U))\}$. All machines work without idle times: machine 1 is utilized in interval $(0, 3E]$, machine 2 works in interval $(0, 4E]$ and machine 3 is busy in $(E, 4E]$.

- (2) Let $\{\pi^l\}_{l \in \mathcal{M}}$ be the family of permutations, for which job prohibitions is taken into account and condition $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$ are satisfied.

From the feasibility we have starting and completion times $t^{st}(2n_0 + 2, 2) = 0, t^f(2n_0 + 2, 2) = 2E, t^{st}(2n_0 + 1, 2) = 2E, t^f(2n_0 + 1, 2) = 4E$ on the second machine. So, $t^f(2n_0 + 1, 1) \leq 2E$ and $t^{st}(2n_0 + 2, 3) \geq 2E$.

Denote by N_1 the subset of jobs, placed in permutation π^1 before job $2n_0 + 1$, and N_2 be the subset of jobs from $\{1, \dots, 2n_0\}$, started before time $2E$ on machine 1, but placed in π^1 after job $2n_0 + 1$. Let $N'_1 \subseteq N_1$ and $N'_2 \subseteq N_2$ be the subset of jobs, started before time $2E$ on machine 3 in the schedule, corresponded permutation π^3 .

Suppose that the Ordered Partition Problem has a negative answer, then $\sum_{i \in N_1} t_{i,1} = \sum_{i \in N_1} t_{i,3} = \sum_{i \in N_1} e_i < E$.

If $N'_1 = N_1$, then $N'_2 = \emptyset$ ($\{\pi^l\}_{l \in \mathcal{M}}$ satisfy job prohibition constraints) and time budget $2E$ is not sufficient for executing the rest jobs in interval $(2E, 4E]$. As a result we go to the contradiction with condition $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$.

Similarly, if $N'_1 \subset N_1$, then $N'_2 \subseteq N_2$, the total duration of all jobs from $N_1 \cup N_2$ on machine 1 does not exceed E , and the total duration of jobs from $N'_1 \cup N'_2$ on machine 3 is less than E . Therefore, after the moment $2E$ the sum of processing times of the rest jobs on machine 3 is greater than $2E$, and we also go to the contradiction with $C_{\max}(\{\pi^l\}_{l \in \mathcal{M}}) \leq 4E$. The presented transformation is linear in n_0 . So, problem $J2||C_{\max}$ with job prohibitions is NP-hard.

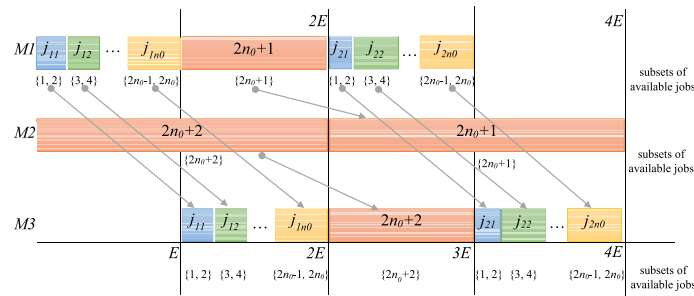


Fig. 8 Illustrative example to the reduction of the Ordered Partition Problem to $F3|Miss - Oper|C_{\max}$ with job prohibitions. The arrows indicate precedence relations between jobs. Here we present a scheme of a non-idle schedule, where job $(2n_0 + 1)$ is executed in interval $[E, 4E)$, job $(2n_0 + 2)$ is performed in interval $[0, 3E)$ on the appropriate machines. Jobs from the set $\{1, 2, \dots, 2n_0\}$ are divided into two

parts $J_1 = \{j_{11}, \dots, j_{1n_0}\}$ and $J_2 = \{j_{21}, \dots, j_{2n_0}\}$ of the same total duration and executed on the first and third machines sequentially without idle times. Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only such schedules have objective $C_{\max} \leq 4E$

The reduction of the Ordered 2-Partition Problem to the decision version of $F3|Miss - Oper|C_{\max}$ is constructed in $O(n_0)$ operations, and therefore, the considered problem is NP-hard.

□

If we have $d_j = 4E$, $j \in \mathcal{J}$, in problem $F|Miss - Oper|\gamma$, $\gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, then Theorem 13 guarantees that

Theorem 14 *Flow-shop scheduling problem $F3|d_j = d, Miss - Oper|\gamma$, $\gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions is NP-hard even for a system of subsets $X = \{X^{il}\}$, such that $|X^{il}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

The flow-shop scheduling problems $F3||C_{\max}$, $F2|d_j|\gamma$, $\gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions and without missing operations are also NP-hard. This result can be obtained from Kovalenko (2016).

Now we show that the flow-shop scheduling problem with job prohibitions and without missing operations is NP-hard in the case of two machines, when the total completion time is minimized.

Theorem 15 *Flow-shop scheduling problem $F2||\sum_j C_j$ with job prohibitions is NP-hard even for a system of subsets $X = \{X^{il}\}$, where $|X^{il}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

Proof Put $m = 2$, $n = 2n_0 + 3$, and processing times

$$\begin{aligned} p_{1,j} &= 1, \quad p_{2,j} = e_j, \quad j = 1, \dots, 2n_0; \\ p_{1,2n_0+1} &= E, \quad p_{2,2n_0+1} = E; \\ p_{1,2n_0+2} &= 2E - n_0, \quad p_{2,2n_0+2} = E; \\ p_{1,2n_0+3} &= 2E - n_0, \quad p_{2,2n_0+3} = E. \end{aligned}$$

As we can see all jobs are executed on both machines.

Set the system of job subsets for positions as

$$X^{i+n_0+2,1} = X^{i+1,1} = \{2i-1, 2i\}, \quad i = 1, \dots, n_0, \quad (23)$$

$$X^{1,1} = \{2n_0+1\}, \quad X^{n_0+2,1} = \{2n_0+2\},$$

$$X^{2n_0+3,1} = \{2n_0+3\}. \quad (24)$$

$$X^{i,2} = X^{i,1}, \quad i = 1, \dots, 2n_0+3. \quad (25)$$

We claim that family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$ satisfying job prohibition constraints and $\sum_j C_j(\{\pi^l\}_{l \in \mathcal{M}}) \leq L := \sum_{j=1}^{n_0} (n_0 - j + 1)(e_{2j-1} + e_{2j}) + 2E + 4E + 6E + 2En_0 + 4En_0$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 9).

- (1) Let the Ordered Partition Problem has a positive answer with subsets A_1, A_2 of set A such that $\sum_{a_i \in A_1} e_i = \sum_{a_i \in A_2} e_i = E$ and $|A_1| = |A_2| = n_0$. We form permutations of jobs $J_1 = (j'_1, \dots, j'_{n_0})$ and $J_2 = (j''_1, \dots, j''_{n_0})$, corresponding to elements from A_1 and A_2 , respectively. Here $j'_i < j'_{i+1}$ and $j''_i < j''_{i+1}$ for all $i = 1, \dots, n_0 - 1$. Then permutations $\{\pi'^1 = (2n_0+1, J_1, 2n_0+2, J_2, 2n_0+3), \pi'^2 = \pi'^1\}$ place jobs according to the system of constraints (23)–(25) and condition $\sum_j C_j(\{\pi^l\}_{l \in \mathcal{M}}) = L$.
- (2) Consider a family of permutations $\{\pi^l\}_{l \in \mathcal{M}}$, that places jobs according to the system of constraints (23)–(25) and has $\sum_j C_j(\{\pi^l\}_{l \in \mathcal{M}}) \leq L$.

According to prohibition constraint job $2n_0+1$ ($2n_0+2$) is performed in position 1 (n_0+2), and the last position contains job $2n_0+3$ on both machines. Denote by J_i , $i = 1, 2$, the subset of jobs performed between jobs $2n_0+i$ and $2n_0+i+1$ on the second machine ($|J_1| = |J_2| = n_0$). It is required to prove that the total duration of jobs from J_1 is equal to E .

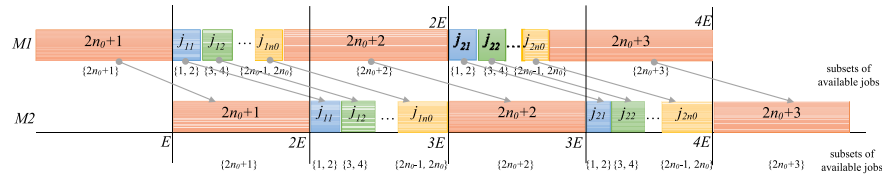


Fig. 9 Illustrative example to the reduction of the Ordered Partition Problem to $F2||\sum_j C_j$ with job prohibitions. Here we present a scheme of a non-idle schedule, where job $(2n_0 + 1)$ is executed in interval $[0, 2E)$, job $(2n_0 + 2)$ is executed in interval $[E + n_0, 4E)$ and job $(2n_0 + 3)$ is performed in interval $[3E + n_0, 6E)$ on both machines in total. Jobs from the set $\{1, 2, \dots, 2n_0\}$ are divided into two parts of the

Suppose that $\sum_{j \in J_1} p_{2,j} < E$, then $\sum_{j \in J_2} p_{2,j} > E$ and $\sum_j C_j(\{\pi^l\}_{l \in M}) \geq L + 1$ as job $2n_0 + 3$ is completed at time $6E + 1$ or later. If the total duration of jobs from J_1 is greater than E , then $\sum_j C_j(\{\pi^l\}_{l \in M}) \geq L + 1 + n_0$ since job $2n_0 + 2$ completes not early than time moment $(4E + 1)$, and as a result jobs from J_2 start execution after time moment $(4E + 1)$.

Therefore, $\sum_{j \in J_1} p_{2,j} = \sum_{j \in J_2} p_{2,j} = E$ and the Ordered Partition Problem has a positive answer. $\square$

3.2 Open-shop scheduling with job prohibitions

In the open-shop scheduling problem the order in which the jobs are processed on the machine, and the order in which the job is processed by the machines can be chosen arbitrarily. In the statement with job prohibitions we have additional constraints, indicating that position i of machine l may contain only jobs from the given subset $X^{i,l} \subseteq \mathcal{J}$.

If we suggest that $X^{i,l} = \{i\}$, i.e. $|X^{i,l}| = 1$, for all $l \in \mathcal{M}$ and reverse the sense of jobs and machines. Then we obtain the classic flow-shop scheduling problem, which is NP-hard even in the case of three jobs (Sotskov & Shakhlevich, 1995). So, the open-shop problem with job prohibitions is NP-hard even in the case of three machines. Moreover, using the same arguments as in the Theorems 11, 12 for job-shop scheduling with the system of job subsets of the cardinality at most two, and two machines we can prove that

Theorem 16 *Open-shop scheduling problem $O2|Miss - Oper|\gamma$, $\gamma \in \{C_{\max}, L_{\max}, \sum_j U_j, \sum_j T_j\}$, with job prohibitions is NP-hard even for a system of subsets $X = \{X^{i,l}\}$, where $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

Here we have no precedence relations between operations in contrast to the job-shop scheduling, but prohibitions used in proofs of Theorems 11, 12 and threshold values define such relations.

Theorem 17 *Open-shop scheduling problem $O2|Miss - Oper|\sum_j C_j$ with job prohibitions is NP-hard even for a*

same total duration $E + n_0$ and performed in $[E, E + n_0) \cup [2E, 3E)$ and $[3E, 3E + n_0) \cup [4E, 5E)$, respectively. Such schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form have objective $\sum_j C_j \leq L$

system of subsets $X = \{X^{i,l}\}$, where $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.

Proof Put $m = 2$, $n = 2n_0 + 3$, and processing times

$$\begin{aligned} p_{1,j} &= 0, \quad p_{2,j} = e_j, \quad j = 1, \dots, 2n_0; \\ p_{1,2n_0+1} &= 2E, \quad p_{2,2n_0+1} = E; \\ p_{2,2n_0+2} &= 2E, \quad p_{1,2n_0+2} = E; \\ p_{1,2n_0+3} &= 0, \quad p_{2,2n_0+3} = E. \end{aligned}$$

Jobs $j = 1, \dots, 2n_0$ and $2n_0 + 3$ consist of one operation, performed on the second machine. Jobs $2n_0 + 1$ and $2n_0 + 2$ consist of two operations, one of which requires the first machine and the other job uses the second machine.

Set the system of job subsets for positions as

$$X^{1,1} = \{2n_0 + 1\}, \quad X^{2,1} = \{2n_0 + 2\}, \quad (26)$$

$$X^{i+n_0+2,2} = X^{i,2} = \{2i - 1, 2i\}, \quad i = 1, \dots, n_0, \quad (27)$$

$$\begin{aligned} X^{n_0+1,2} &= \{2n_0 + 2\}, \quad X^{n_0+2,2} = \{2n_0 + 1\}, \\ X^{2n_0+3,2} &= \{2n_0 + 3\}. \end{aligned} \quad (28)$$

We claim that family of permutations $\{\pi^l\}_{l \in M}$ satisfying the job prohibition constraints and $\sum_j C_j(\{\pi^l\}_{l \in M}) \leq L := \sum_{j=1}^{n_0} (n_0 - j + 1)(e_{2j-1} + e_{2j}) + 4E + 3E + 3En_0 + 5E$ exists if and only if the Ordered Partition Problem has a positive answer (see Fig. 10).

- (1) Let the Ordered Partition Problem has a positive answer with subsets A_1, A_2 of set A such that $\sum_{a_i \in A_1} e_i =$

$$\sum_{a_i \in A_2} e_i = E \text{ and } |A_1| = |A_2| = n_0.$$

We form permutations of jobs $J_1 = (j'_1, \dots, j'_{n_0})$ and $J_2 = (j''_1, \dots, j''_{n_0})$, corresponding to elements from A_1 and A_2 , respectively. Here $j'_i < j'_{i+1}$ and $j''_i < j''_{i+1}$ for all $i = 1, \dots, n_0 - 1$. Then permutations $\{\pi^1 = (2n_0 + 1, 2n_0 + 2), \pi^2 = (J_1, 2n_0 + 2, 2n_0 + 1, J_2, 2n_0 + 3)\}$ satisfy the system of constraints (26)–(28) and condition $\sum_j C_j(\{\pi^l\}_{l \in M}) = L$.

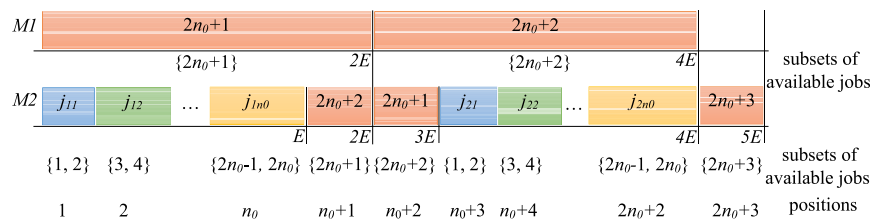


Fig. 10 Illustrative example to the reduction of the Ordered Partition Problem to $O2||\sum_j C_j$ with job prohibitions. Here we present a scheme of a non-idle schedule, where job $(2n_0 + 1)$ is executed in interval $[0, 2E)$ on the first machine and in $[2E, 3E)$ on the second machine, job $(2n_0 + 2)$ is executed in interval $[E, 2E)$ on the second machine and in $[2E, 4E)$ on the first machine, and job $(2n_0 + 3)$ is performed in interval $[4E, 5E)$ on the second machine.

- (2) Consider a family of permutations $\{\pi^l\}_{l \in M}$, that satisfies the system of constraints (26)–(28) and has $\sum_j C_j(\{\pi^l\}_{l \in M}) \leq L$.

According to prohibition constraints job $2n_0 + 2$ ($2n_0 + 1$) is performed in position $n_0 + 1$ ($n_0 + 2$), and the last position contains job $2n_0 + 3$ on the second machine. Denote by J_1 the subset of jobs performed before job $2n_0 + 2$ and by J_2 the subset of jobs executed after job $2n_0 + 1$ and before job $2n_0 + 3$ on the second machine ($|J_1| = |J_2| = n_0$). It is required to prove that the total duration of jobs from J_1 is equal to E . Suppose that $\sum_{j \in J_1} p_j < E$, then $\sum_{j \in J_2} p_j > E$ and $\sum_j C_j(\{\pi^l\}_{l \in M}) \geq L + 1$ as job $2n_0 + 3$ is completed at time $5E + 1$ or later. If the total duration of jobs from J_1 is greater than E , then $\sum_j C_j(\{\pi^l\}_{l \in M}) \geq L + 2 + n_0$ since jobs from J_2 start execution after time moment $(3E + 1)$, job $2n_0 + 1$ completes not early than time moment $(3E + 1)$, and job $2n_0 + 2$ completes at time $4E + 1$ or later.

Therefore, $\sum_{j \in J_1} p_j = \sum_{j \in J_2} p_j = E$ and the Ordered Partition Problem has a positive answer. $\square$

3.3 Solving approach

Enumerating all possible combinations of feasible permutations of jobs consistent with job prohibitions on machines independently and using the approach from Sect. 2.2 we obtain the following exact algorithm.

For each machine l (independently of the others), a bipartite graph is constructed, its special edges and cycles are computed in $O(n_l)$ time. Next, all possible combinations of perfect matchings in the graphs of machines are enumerated. The total number of combinations is $O(1.42^{n_{\max} m})$. For each combination, an early feasible schedule is constructed and the objective function is calculated in $O(n_{\max} m + k_{\max} n)$

The rest jobs are divided into two parts $J_1 = \{j_{11}, \dots, j_{1n_0}\}$ and $J_2 = \{j_{21}, \dots, j_{2n_0}\}$ of the same total duration E and performed in intervals $[0, E)$ and $[3E, 4E)$ on the second machine. Such a schedule can be constructed if and only if the corresponding Ordered Partition Instance has a positive answer. Only schedules of this form have objective $\sum_j C_j \leq \sum_{j=1}^{n_0} (n_0 - j + 1)(e_{2j-1} + e_{2j}) + (3n_0 + 12)E$

time. The solution with the best value of the objective function corresponds to the optimal schedule. Thus, we have

Theorem 18 *Job-shop problem $J||\gamma$, $\gamma \in \{C_{\max}, L_{\max}, \sum_j T_j, \sum_j C_j, \sum_j U_j\}$, with job prohibitions can be solved in $O(1.42^{n_{\max} m}(nk_{\max} + mn_{\max}))$ time for a system of job subsets $X^{i,l}$ such that $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

From Theorem 10 we conclude that almost all systems of job subsets $X^{i,l}$ such that $|X^{i,l}| \leq 2$ produce polynomially solvable instances of $Jm||\gamma$, $Fm||\gamma$, $Fm|Miss - Oper||\gamma$, $\gamma \in \{C_{\max}, L_{\max}, \sum_j T_j, \sum_j C_j, \sum_j U_j\}$ with job prohibitions.

For arbitrary systems of job subsets on machines (without bounds on the cardinality) a feasible schedule can be found in polynomial time by solving assignment instances (Munkres, 1957) for all m machines independently. The optimal solution in this case may be computed in $O(n_{\max}^{2.5} m + (n_{\max}^{pm})^m (mn_{\max}^2 + nk_{\max}))$ time by configuring perfect matchings (see Sect. 2.2) in the corresponding bipartite graphs of machines ($n_{\max}^{pm}$ is the maximal number of perfect matchings in graphs of machines) and computing objectives for appropriate schedules.

In the case of open-shop scheduling problem with the makespan criterion we can apply such approach for solving with enumerating of permutations of operations on machines only for the case $m = 2$. Here we obtain the classical polynomially solvable instances of the job-shop problem for computing the optimal objective under the given sequences on two machines (Tanaev et al., 1989) (sequences present two jobs with the given routes of operations for the job-shop). The objective function is computed in $O(n \log(n))$ time (Tanaev et al., 1989) for the given permutation. So we have

Theorem 19 *Open-shop problem $O2||C_{\max}$ with job prohibitions can be solved in $O(1.42^{2n} n \log(n))$ time for a system of job subsets $X^{i,l}$ such that $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.*

Recall that $O2||C_{\max}$ without job prohibitions is polynomially solvable (Tanaev et al., 1989).

The objective cannot be computed in polynomial time for the fixed sequences of jobs on three and more machines if $P \neq NP$ (here we obtain the NP-hard job-shop problem (Tanaev et al., 1989)).

3.4 Integer programming models

Now we present an adaptation of the classic mixed-integer formulation based on position representation (Ku & Beck, 2016) for the job-shop and flow-shop problems with job prohibitions.

Let O be the set of all operations of all jobs, and O_l be the subset of operations, that must be performed on machine $l \in \mathcal{M}$, $n_l := |O_l|$. We denote by p_o the duration of operation o , and by $Y^{o,l}$ the subset of positions, where operation o may be placed on machine l , where $o = O_v^j$, $i = L_{vj}$, $j \in \mathcal{J}$, $v = 1, \dots, k_j$. Let $n_{\max} := \max_{l \in \mathcal{M}} n_l$ and $k_{\max} := \max_{j \in \mathcal{J}} k_j$.

Introduce the following Boolean variables:

$$x_{oi} = \begin{cases} 1, & \text{if operation } o \text{ is performed in position } i \text{ on machine } l, \\ 0 & \text{otherwise; } i = 1, \dots, n_l, \ o \in X^{i,l}, \ l \in \mathcal{M}, \end{cases}$$

and continuous variables $C_{li} \geq 0$ (the completion time of a job operation in position $i = 1, \dots, n_l$ of machine $l \in \mathcal{M}$).

Then the set of feasible solutions is defined as follows:

$$\sum_{o \in X^{i,l}} x_{oi} = 1, \ l \in \mathcal{M}, \ i = 1, \dots, n_l, \quad (29)$$

$$\sum_{i \in Y^{o,l}} x_{oi} = 1, \ l \in \mathcal{M}, \ o \in O_l, \quad (30)$$

$$C_{li} + \sum_{o \in X^{i+1,l}} p_o x_{oi+1} \leq C_{l,i+1}, \ l \in \mathcal{M}, \quad (31)$$

$$i = 1, \dots, n_l - 1,$$

$$C_{l_1,i_1} + p_{o_2} \leq C_{l_2,i_2} + \quad (32)$$

$$M(1 - x_{o_1,i_1}) + M(1 - x_{o_2,i_2}),$$

$$j \in \mathcal{J}, \ v = 1, \dots, k_j - 1, \ o_1 = O_v^j, \ o_2 = O_{v+1}^j,$$

$$l_1 = L_{v,j}, \ l_2 = L_{v+1,j}, \ i_1 \in Y^{o_1,l_1}, \ i_2 \in Y^{o_2,l_2},$$

$$C_{li} \geq 0, \ x_{oi} \in \{0, 1\}, \ o \in X^{i,l}, \ l \in \mathcal{M},$$

$$i = 1, \dots, n_l. \quad (33)$$

Constraint (29) implies that exactly one allowable operation is assigned in each position. Conditions (30) guarantee that all operations of jobs are placed in positions. Inequality (31) provides the lower bound on the completion time of an operation in the next position of a machine. Conditions (32) ensure to satisfaction of the partial order between operations of the same job. Constraint (33) give the definition of the feasible region of variables. Here we have $O(mn_{\max})$

Boolean variables and $O(mn_{\max})$ continues variables for the system of job subsets for positions with $|X^{i,l}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$.

Criteria can be written as: the makespan

$$C_{\max} \rightarrow \min,$$

$$C_{l,n_l} \leq C_{\max}, \ l \in \mathcal{M},$$

the maximum lateness

$$L_{\max} \rightarrow \min,$$

$$L_{\max} \geq C_{li} - \sum_{o \in X^{i,l}, \ j \in \mathcal{J}: o = O_v^j} x_{oi} d_j, \ l \in \mathcal{M},$$

$$i = 1, \dots, n_l,$$

the total tardiness

$$\sum_{j=1}^n T_j \rightarrow \min,$$

$$T_j \geq 0, \ T_j \geq C_{li} - d_j - M(1 - x_{oi}), \ j = 1, \dots, n,$$

$$o = O_{k_j}^j, \ k \in Y^{o,L_{k_j,j}},$$

the total completion time

$$\sum_{j=1}^n C_j \rightarrow \min,$$

$$C_j \geq C_{li} - M(1 - x_{oi}), \ j = 1, \dots, n,$$

$$o = O_{k_j}^j, \ i \in Y^{o,L_{k_j,j}},$$

the number of tardy jobs

$$\sum_{j=1}^n U_j \rightarrow \min,$$

$$C_j \leq d_j + U_j \cdot M, \ U_j \in \{0, 1\}, \ j = 1, \dots, n.$$

Now we present the model based on matchings of cycles for the case $|X^{il}| \leq 2$, $l \in \mathcal{M}$, $i = 1, \dots, n_l$, similar to the model for the single-processor case from Sect. 2.3. Let $s = 1, \dots, q(l)$ enumerate cycles of the bipartite graphs corresponding to the systems of job subsets on machines $l \in \mathcal{M}$.

Consider the following Boolean variables:

$$x_{sl} = \begin{cases} 0, & \text{if the first matching is selected in cycle } s \text{ of machine } l, \\ 1, & \text{if the second matching is selected in cycle } s \text{ of machine } l; \end{cases}$$

and continuous variables $C_{li} \geq 0$ (the completion time of a job operation in position i of machine l), $C(O_v^j) \geq 0$ (the completion time of operation O_v^j).

We will use the following notations:

s_{li} is the cycle of machine l , containing position i ;
 s_{vj} is the cycle of machine L_{vj} , containing operation O_v^j ;
 $p_{li}^0(p_{li}^1)$ is the duration of the operation in position i of machine l , corresponding to the first (the second) matching of cycle s_{li} ;
 $k_{vj}^0(k_{vj}^1)$ is the position of machine L_{vj} , corresponding to operation O_v^j in accordance with the first (the second) matching of cycle l_{vj} .

In this approach the constraints are written in the form:

$$C_{li} + p_{li}^0(1 - x_{s_{li},l}) + p_{li}^1 x_{s_{li},l} \leq C_{l,i+1}, \quad l \in \mathcal{M}, \quad i = 1, \dots, n_l - 1, \quad (34)$$

$$C(O_v^j) \geq C_{L_{vj},k_{vj}^0} - M(x_{s_{vj},L_{vj}}), \quad j \in \mathcal{J}, \quad v = 1, \dots, k_j - 1, \quad (35)$$

$$C(O_v^j) \geq C_{L_{vj},k_{vj}^1} - M(1 - x_{s_{vj},L_{vj}}), \quad j \in \mathcal{J}, \quad v = 1, \dots, k_j - 1, \quad (36)$$

$$C(O_v^j) + p_{v+1,j} \leq C(O_{v+1}^j), \quad j \in \mathcal{J}, \quad v = 1, \dots, k_j - 1, \quad (37)$$

$$C(O_v^j) \geq 0, \quad j \in \mathcal{J}, \quad v = 1, \dots, k_j - 1, \quad (38)$$

$$C_{li} \geq 0, \quad l \in \mathcal{M}, \quad i = 1, \dots, n_l, \quad (39)$$

$$x_{sl} \in \{0, 1\}, \quad l \in \mathcal{M}, \quad s = 1, \dots, q(l). \quad (40)$$

Constraint (34) gives the relation between completion times of operations in consecutive positions. Conditions (30) define completion time of operations through completion times in positions. Inequality (31) guarantees the satisfaction of precedence constraints between operations. Constraints (38)–(40) give the area where the variables are defined. Criteria can be formulated as in the previous case.

The model (34)–(40) contains $O(mq_{\max})$ Boolean variables and $O(mn_{\max} + nk_{\max})$ continuous variables, where $q_{\max} = \max_{l \in \mathcal{M}} q(l)$. Note that the number of Boolean variables will be sufficiently smaller in contrast to the previous model for small number of cycles in bipartite graphs of machines.

For the arbitrary multi-machine systems with job prohibitions, the mixed integer programming models can be written in a similar way through position-based or matching-based approaches.

Conclusion

In the current paper we provide the computational complexity analyses of scheduling problems with job prohibitions. NP-hardness proofs based on the reduction of the Ordered Partition Problem are presented for a series of problems

Table 1 NP-hardness of scheduling problems with job prohibitions (plus corresponds to the current paper)

Problem	Cardinality of job subsets	References
<i>Single machine</i>		
$1 \sum_j w_j U_j$	$ X^i = 2$	+
$1 r_j L_{\max}$	$ X^i \leq 2$	+
$1 r_j \sum_j U_j$	$ X^i \leq 2$	+
$1 r_j \sum_j T_j$	$ X^i \leq 2$	+
$1 C_j \leq d_j \sum_j w_j C_j$	$ X^i \leq 2$	+
$1 r_j \sum_j C_j$	$ X^i \leq 2$	+
$1 l_{ij} C_{\max}$	$ X^i \leq 2$	Eremeev and Kovalenko (2012)
$1 \sum_j w_j T_j$	$ X^i \leq 2$	Zakharova (2023a)
<i>Multiple machines</i>		
$2 \gamma \in \{C_{\max}, L_{\max}, \sum_j U_j, \sum_j T_j\}$	$ X^i = 2$	+
$J2 C_{\max}$	$ X^i \leq 2$	+
$J2 \gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$	$ X^i \leq 2$	+
$F3 Miss - Oper C_{\max}$	$ X^i \leq 2$	+
$F3 Miss - Oper \gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$	$ X^i \leq 2$	+
$F3 C_{\max}$	$ X^i \leq 2$	Kovalenko (2016)
$F2 \gamma \in \{L_{\max}, \sum_j U_j, \sum_j T_j\}$	$ X^i \leq 2$	Kovalenko (2016)
$F2 \sum_j C_j$	$ X^i \leq 2$	+
$O3 \gamma \in \{C_{\max}, L_{\max}, \sum_j U_j, \sum_j T_j\}$	$ X^i = 1$	+
$O2 Miss - Oper \gamma \in \{C_{\max}, L_{\max}, \sum_j U_j, \sum_j T_j, \sum_j C_j\}$	$ X^i \leq 2$	+

in single-processor and multi-processor environments (the results are aggregated in Table 1). Moreover, a common enumeration approach and MIP models are given for solving single-stage and multi-stage instances with job prohibitions. We show that the considered scheduling problems with prohibitions are NP-hard in the worst case, but polynomially solvable in “almost all” cases. It is important property for practical applications of evolutionary algorithms and other methods based on iterative solving subproblems with job prohibitions.

Further research can be undertaken to experimental evaluation of various exact and approximate methods. It is also interesting to investigate new properties of cycles in auxiliary bipartite graphs corresponding to the system of available jobs in positions of machines and possibilities of parallelization techniques in enumeration approaches.

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